

A disproof of Conjecture 00000003295

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Dawson's reciprocal expansion

Dawson's integral has the familiar asymptotic expansion

$$F(x) \sim \sum_{n \geq 0} \frac{(2n-1)!!}{2^{n+1}x^{2n+1}} = \frac{1}{2x} + \frac{1}{4x^3} + \frac{3}{8x^5} + \frac{15}{16x^7} + \cdots.$$

The conjecture incorrectly promotes this asymptotic expansion to a series converging throughout the real axis.

Theorem 1. *The displayed series diverges at $x = 1$.*

Proof. Write $a_n = (2n-1)!!/2^{n+1}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{2n+1}{2}.$$

For $n \geq 1$ this ratio is at least $3/2$. In particular $a_4 = 105/32 > 1$ and all subsequent terms increase, so a_n does not tend to zero. A necessary condition for convergence of a series is that its terms tend to zero. Therefore the series diverges at $x = 1$ and cannot have the whole real axis as its convergence domain. $\square$

Indeed the same ratio tends to infinity for every fixed nonzero x , after division by the additional factor x^2 , so the expansion is divergent for every fixed finite nonzero x .

Lean 4 certificate

The project in `lean4/` defines $(2n-1)!!$, proves by induction that $2^{n+1} \leq (2n-1)!!$ for $n \geq 4$, and contradicts the eventual inequality required for terms to approach zero. No `sorry`, `native_decide`, or extra axiom is used.